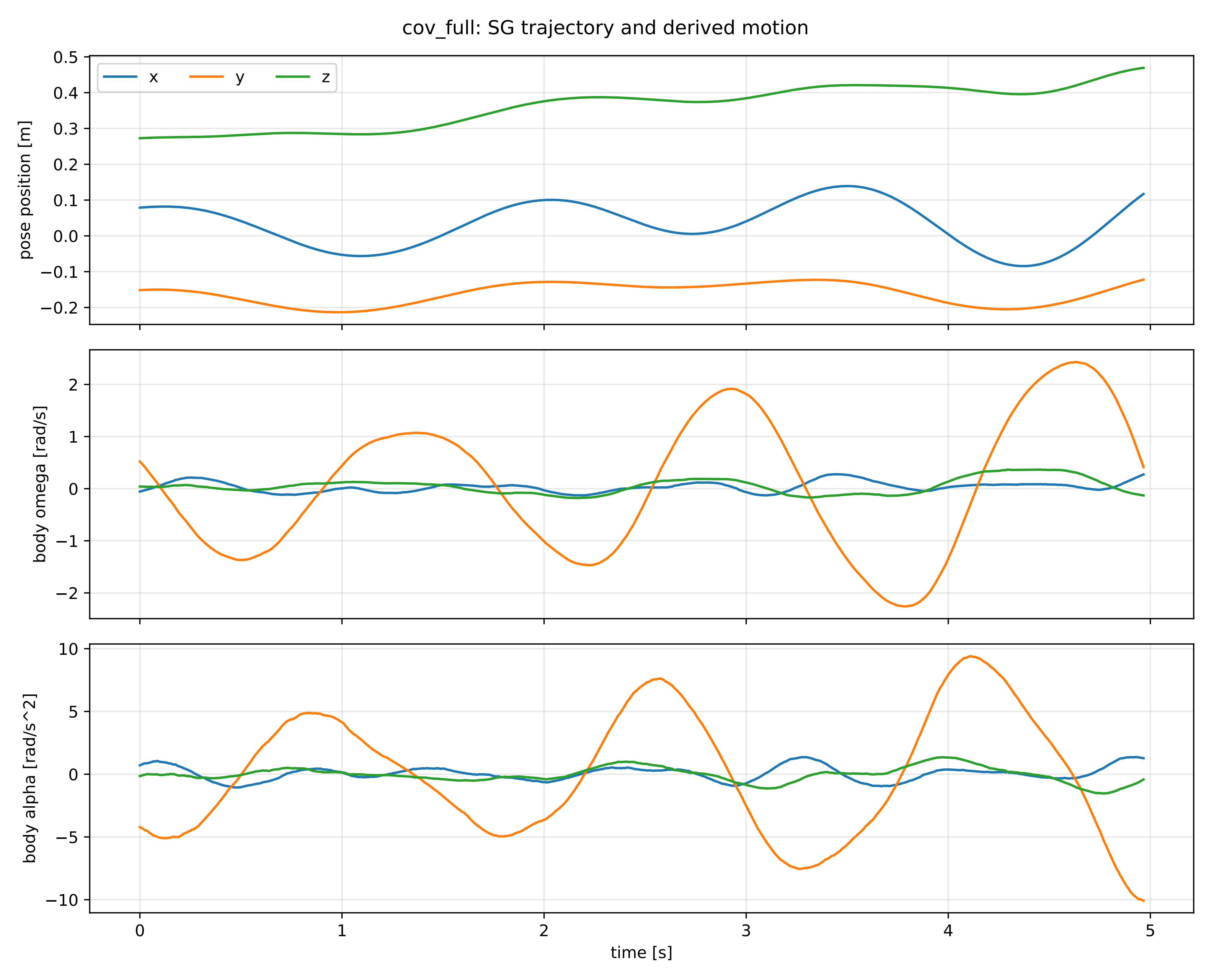
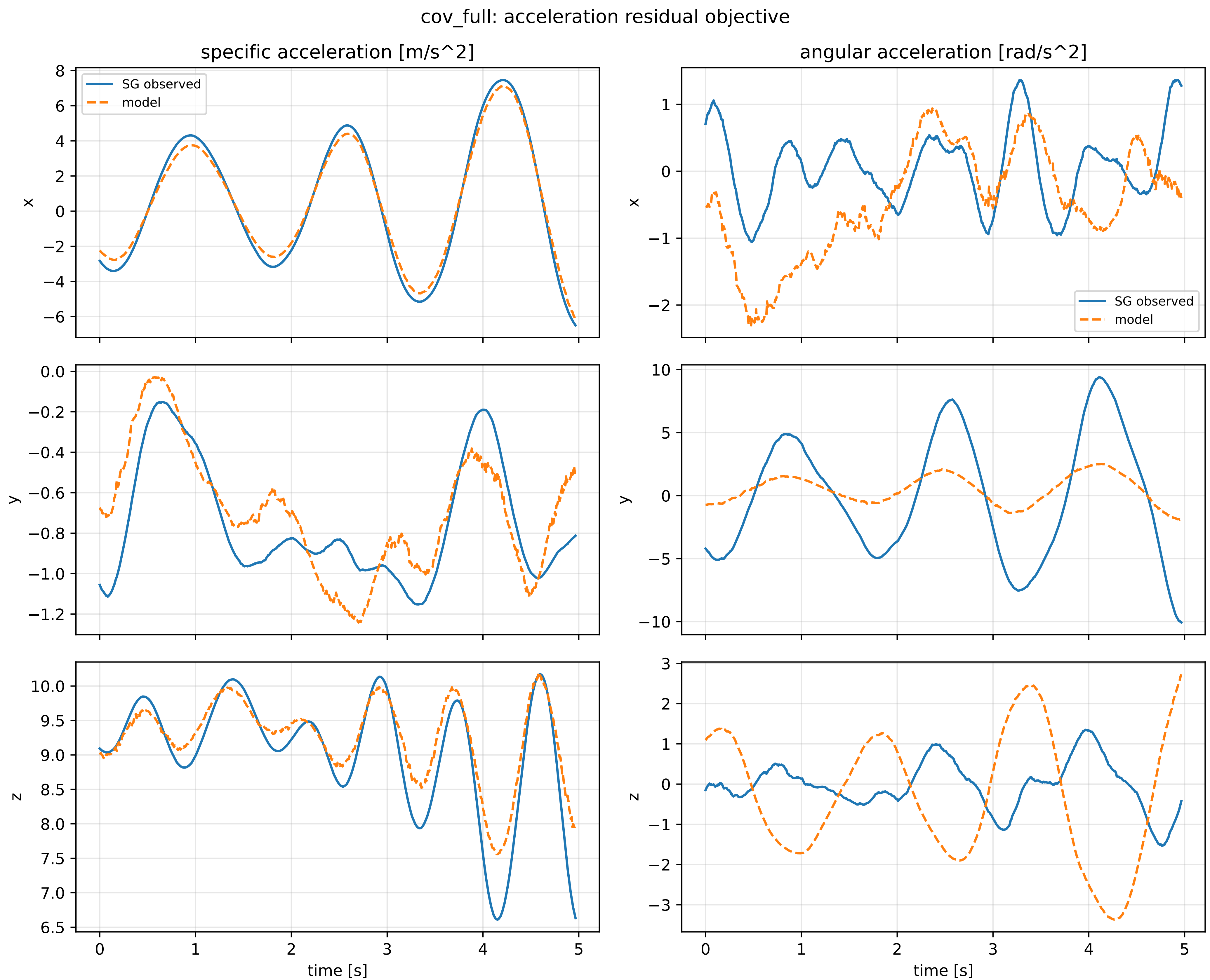
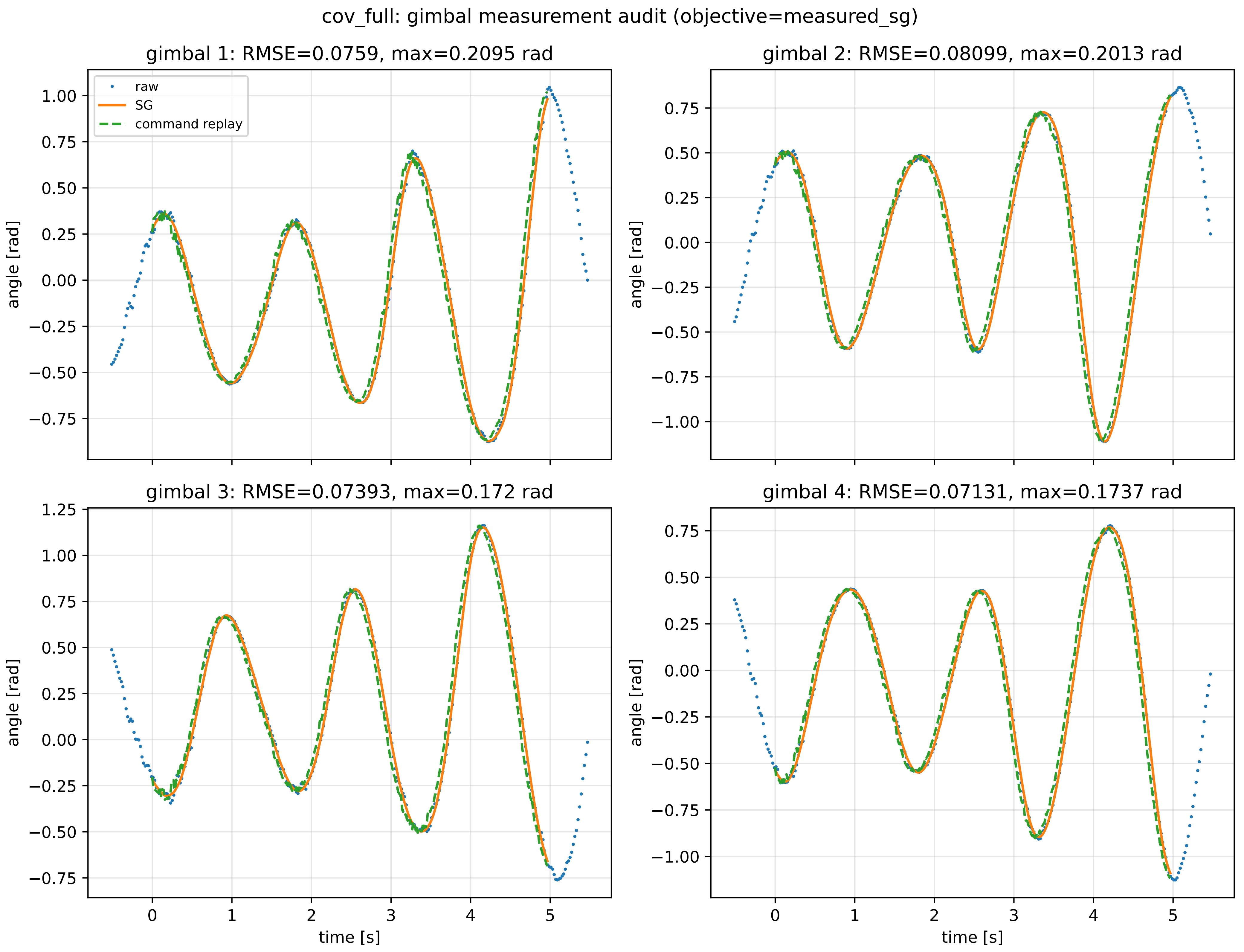
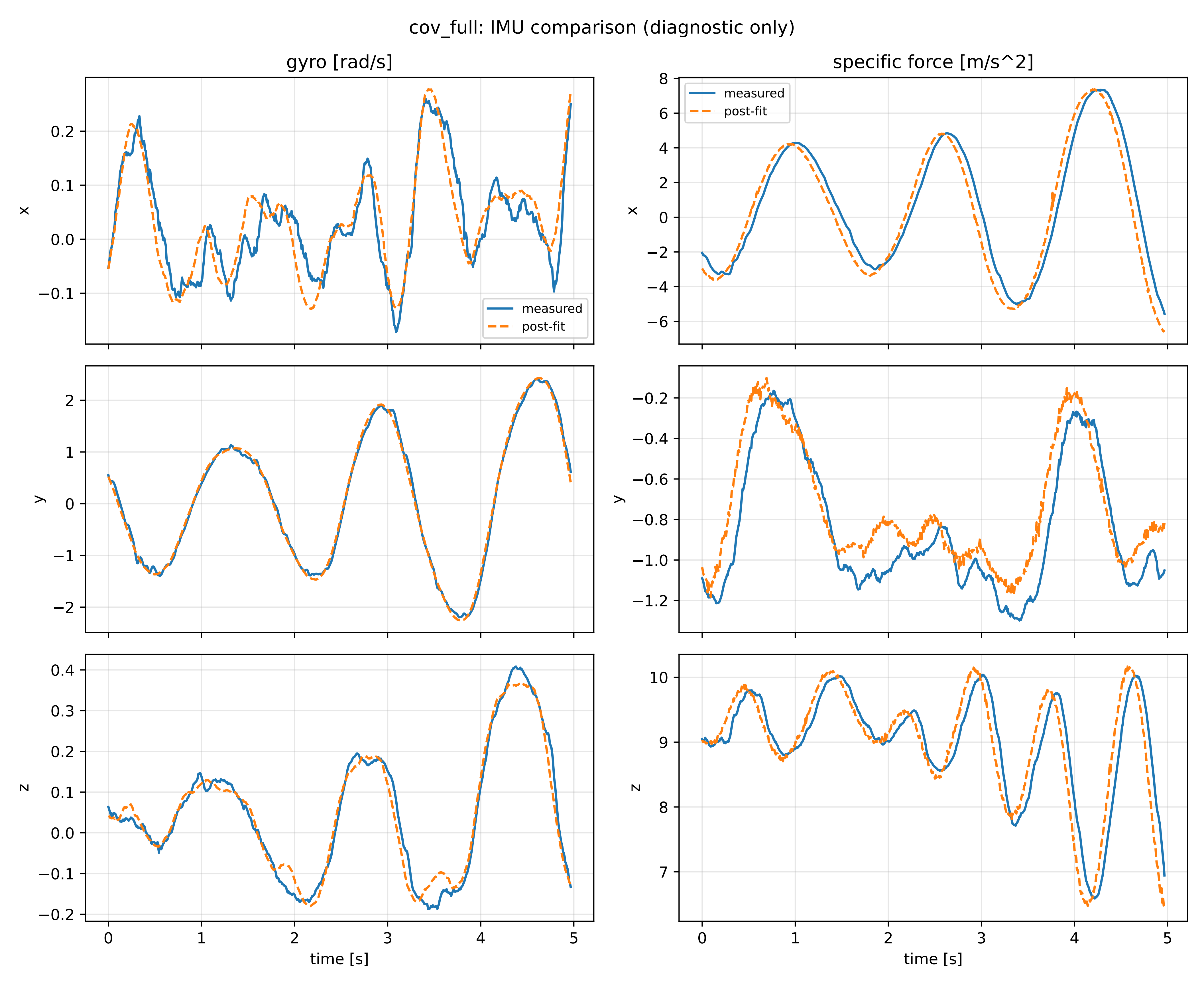
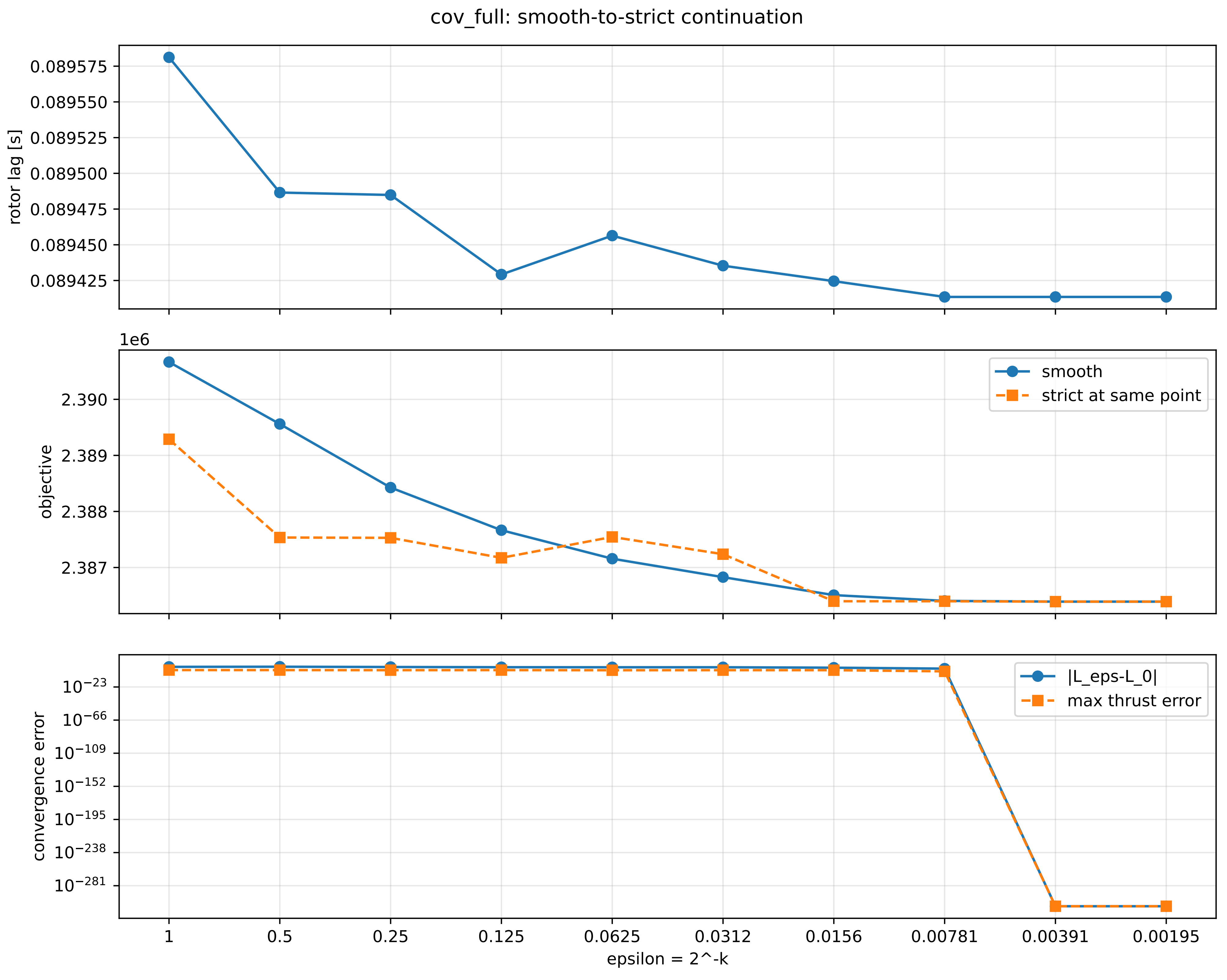

cov_full: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
2
1
0
−1
−2
10
5
0
−5
−10
0
1
2
3
4
5


cov_full: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
SG observed
model
x
y
z
x
y
z
time [s]
time [s]



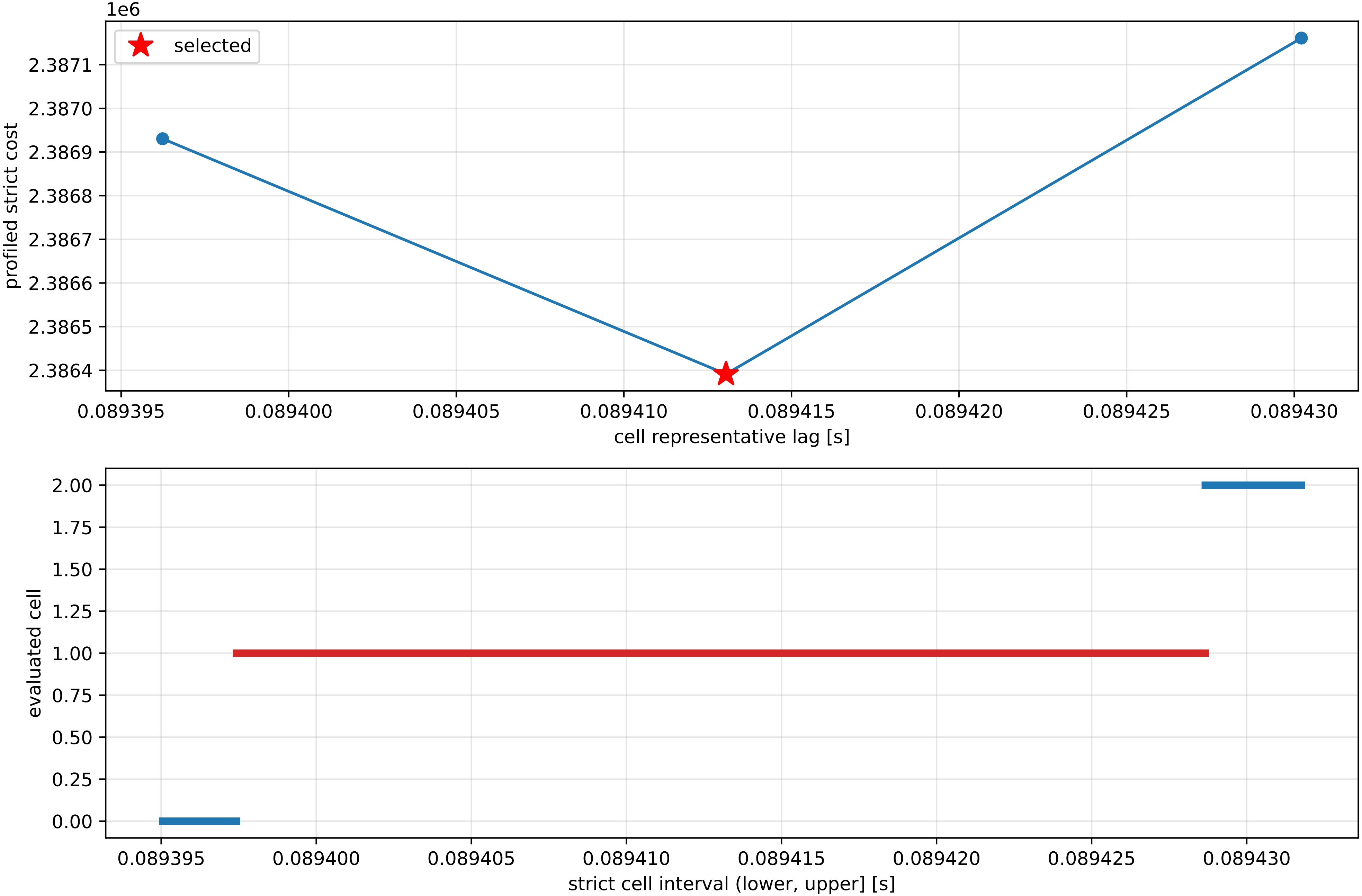
cov_full: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]


cov_full: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



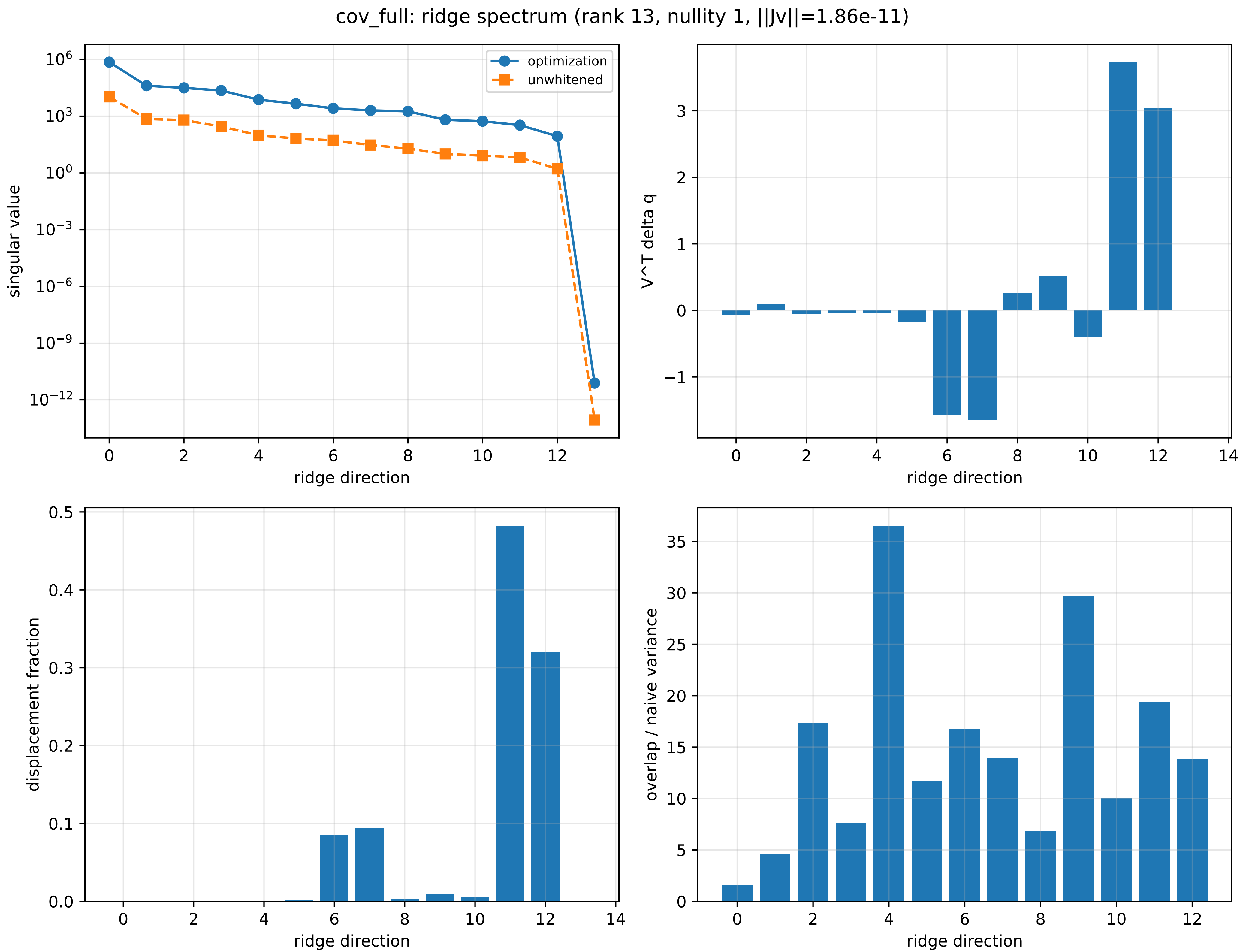
cov_full: smooth-to-strict continuation
rotor lag [s]
0.089575
0.089550
0.089525
0.089500
0.089475
0.089450
0.089425
1e6
objective
2.390
2.389
2.388
2.387
smooth
strict at same point
convergence error
10^{-23}
10^{-66}
10^{-109}
10^{-152}
10^{-195}
10^{-238}
10^{-281}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

cov_full: exact strict-ZOH lag cells

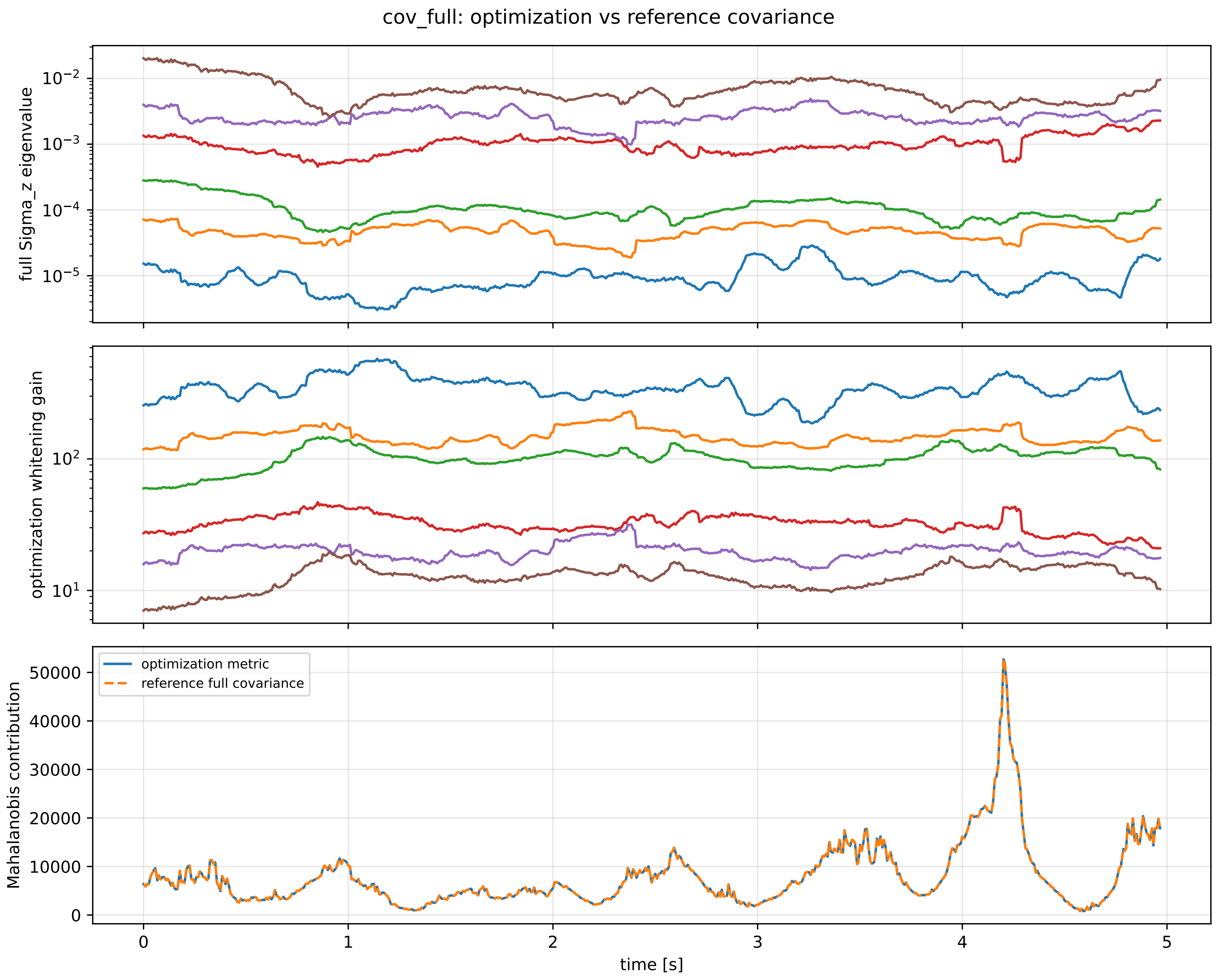


selected (0.0893974304, 0.0894286633] s; final smooth 0.0894134968 s; data boundary=False cov_full: parameter estimate

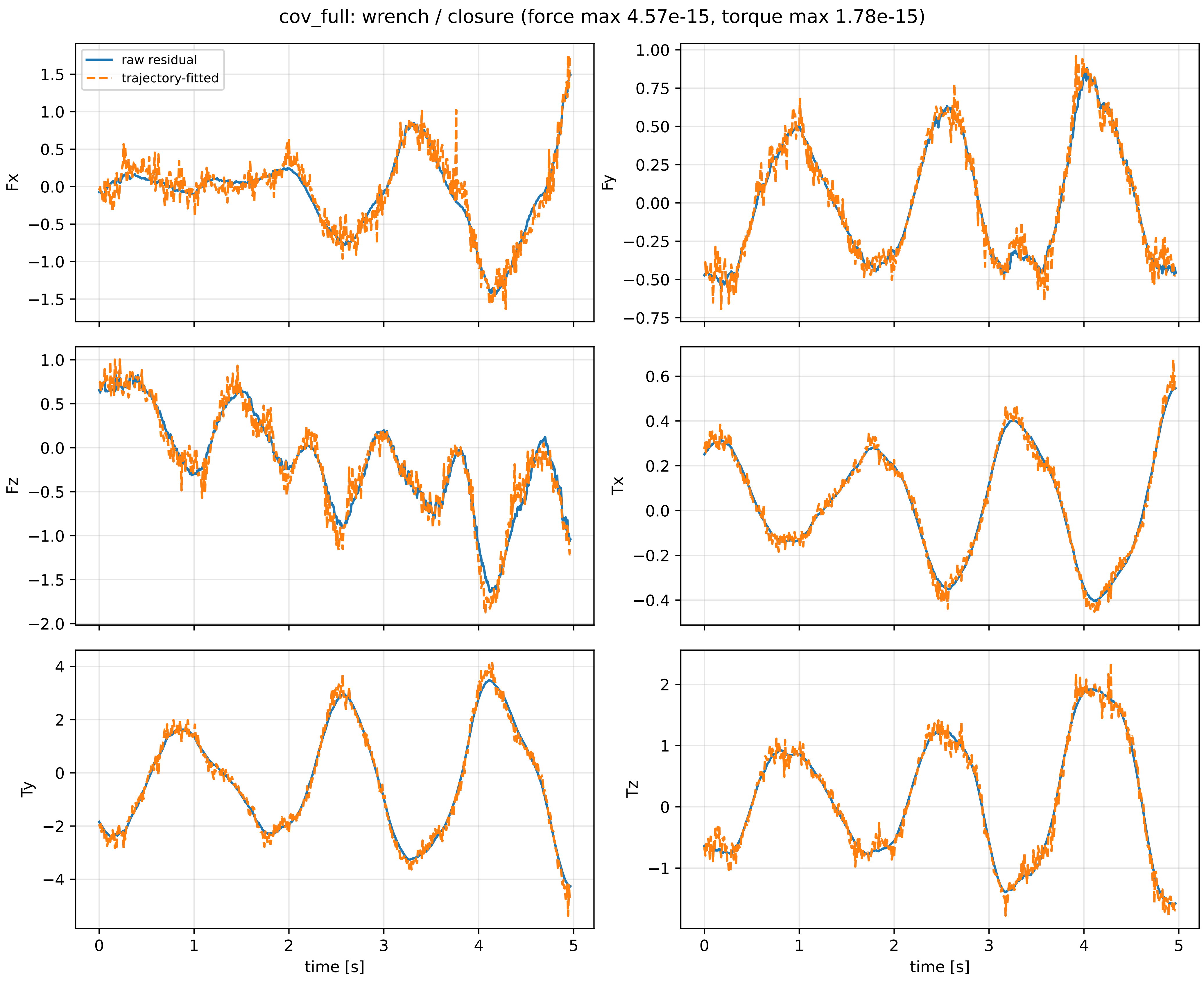
```
Case: cov_full
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.54295976
inertia [kg m^2]:
[[ 0.04094785 -0.04496586 -0.03507921]
 [-0.04496586  0.51812985 -0.01087469]
 [-0.03507921 -0.01087469  0.5104819 ]]
CoG body [m]: [ 0.02143602  0.16267991 -0.0425746 ]
force effectiveness: [0.18439486 0.17365603 0.99957629 0.89685744]

scale-free J/m [m^2]:
[[ 0.02653851 -0.0291426  -0.02273501]
 [-0.0291426   0.33580257 -0.00704794]
 [-0.02273501 -0.00704794  0.33084589]]
scale-free f/m: [0.11950724 0.11254735 0.64783043 0.58125783]

rotor lag cell: (0.0893974304, 0.0894286633] s
representative: 0.0894130468 s
```



cov_full: ridge spectrum (rank 13, nullity 1, ||Jv||=1.86e-11)
optimization
unwhitened
singular value
ridge direction
V^T delta q
displacement fraction
overlap / naive variance



cov_full: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-2}
10^{-3}
10^{-4}
10^{-5}
optimization whitening gain
10^{2}
10^{1}
Mahalanobis contribution
50000
40000
30000
20000
10000
0
optimization metric
reference full covariance
0
1
2
3
4
5
time [s]


cov_full: wrench / closure (force max 4.57e-15, torque max 1.78e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]